

Project Synopsis

NIVRA – AI-Powered Smart Complaint Management System

Introduction

NIVRA is a mobile application being developed using **Flutter** and **Firestore** to provide a modern, efficient, and user-friendly platform for digital complaint registration and grievance management. The project aims to replace traditional complaint handling methods with a centralized digital solution that enables users to register complaints, monitor their status, and access various support services through a secure and intuitive interface.

The application follows a modular architecture and integrates cloud-based technologies to ensure secure authentication, efficient data management, and seamless user experience. By utilizing Firestore services, the application provides reliable user authentication and cloud database support while maintaining scalability for future enhancements.

Objectives

- To develop a digital platform for registering and managing public complaints.
- To provide a secure authentication system for users.
- To enable users to track complaints through a centralized interface.
- To simplify complaint management using cloud-based technologies.
- To build a scalable mobile application that can be extended with AI-powered features and additional services.

Project Overview

The application begins with a **Splash Screen**, followed by **Onboarding Screens** that introduce first-time users to the application's features. The onboarding process is managed using **Shared Preferences**, ensuring that these screens are displayed only once.

A secure authentication system has been implemented using **Firestore Authentication**, allowing users to register and log in using their email and password. User information, including full name, email address, phone number, and unique user ID, is securely stored in **Cloud Firestore**. The application performs input validation and provides appropriate error handling to ensure a smooth registration and login experience.

After successful authentication, users are directed to a personalized **Dashboard**, where they are greeted by name using information retrieved dynamically from Firestore. The dashboard serves as the central hub of the application, providing access to major functionalities such as complaint registration, complaint status tracking, complaint history, profile management, notifications, and application settings.

The **Settings** module provides options for profile access, notification preferences, privacy and security, help and support, application information, and secure logout functionality. Logout is managed through Firebase Authentication while maintaining user preferences such as onboarding completion.

The project follows a service-based architecture where business logic is separated from the user interface, making the application easier to maintain, extend, and scale.

Technologies Used

- Flutter (Dart)
- Firebase Authentication
- Cloud Firestore
- Shared Preferences
- Material Design
- VS Code
- Git & GitHub

Key Features

- Splash Screen
- Interactive Onboarding Screens
- Secure User Registration
- User Login & Authentication
- Forgot Password & Password Recovery
- Personalized User Dashboard
- Dynamic User Greeting
- Complaint Registration
- Complaint Status Tracking
- Complaint History
- Detailed Complaint View
- Profile Management
- Edit Profile
- Notifications & Updates
- Settings & Preferences
- Help & Support
- User Feedback Module
- About NIVRA
- Secure Logout

- Cloud-Based Data Storage using Firebase
- Real-Time Database Synchronization
- User-Friendly Material Design Interface
- Modular and Scalable Application Architecture

Conclusion

NIVRA is designed as a secure, scalable, and user-centric complaint management application that leverages modern mobile development technologies and cloud services. By combining Flutter with Firebase, the project establishes a strong foundation for a comprehensive digital complaint management system that emphasizes usability, security, and maintainability while supporting future expansion as development progresses.